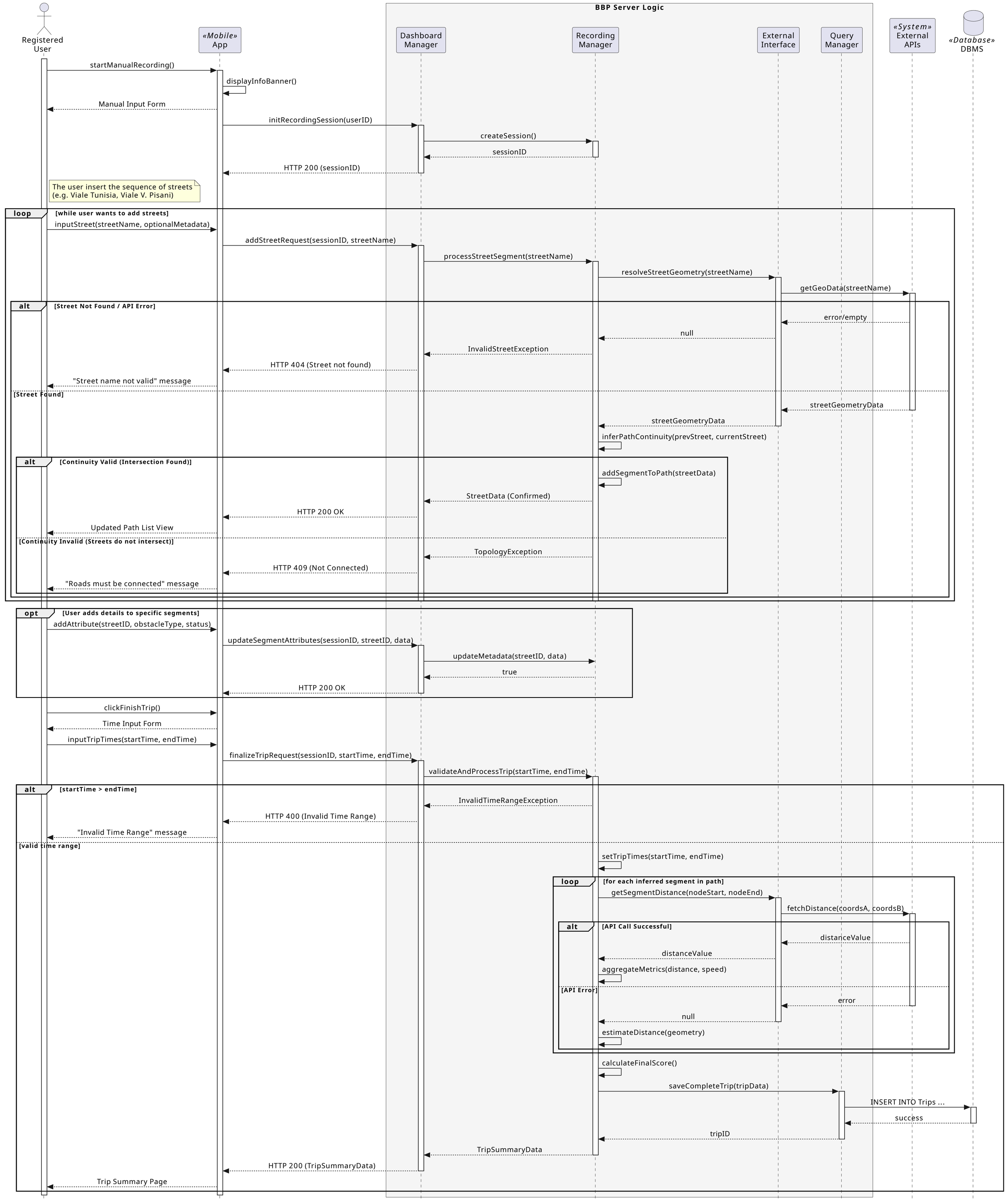
BBP Server Logic
Registered User
«Mobile» App
Dashboard Manager
Recording Manager
External Interface
Query Manager
«System» External APIs
«Database» DBMS
startManualRecording()
displayInfoBanner()
Manual Input Form
initRecordingSession(userID)
createSession()
sessionID
HTTP 200 (sessionID)
The user insert the sequence of streets (e.g. Viale Tunisia, Viale V. Pisani)
loop
[while user wants to add streets]
inputStreet(streetName, optionalMetadata)
addStreetRequest(sessionID, streetName)
processStreetSegment(streetName)
resolveStreetGeometry(streetName)
getGeoData(streetName)
alt
[Street Not Found / API Error]
error/empty
null
InvalidStreetException
HTTP 404 (Street not found)
"Street name not valid" message
[Street Found]
streetGeometryData
streetGeometryData
inferPathContinuity(prevStreet, currentStreet)
alt
[Continuity Valid (Intersection Found)]
addSegmentToPath(streetData)
StreetData (Confirmed)
HTTP 200 OK
Updated Path List View
[Continuity Invalid (Streets do not intersect)]
TopologyException
HTTP 409 (Not Connected)
"Roads must be connected" message
opt
[User adds details to specific segments]
addAttribute(streetID, obstacleType, status)
updateSegmentAttributes(sessionID, streetID, data)
updateMetadata(streetID, data)
true
HTTP 200 OK
clickFinishTrip()
Time Input Form
inputTripTimes(startTime, endTime)
finalizeTripRequest(sessionID, startTime, endTime)
validateAndProcessTrip(startTime, endTime)
alt
[startTime > endTime]
InvalidTimeRangeException
HTTP 400 (Invalid Time Range)
"Invalid Time Range" message
[valid time range]
setTripTimes(startTime, endTime)
loop
[for each inferred segment in path]
getSegmentDistance(nodeStart, nodeEnd)
fetchDistance(coordsA, coordsB)
alt
[API Call Successful]
distanceValue
distanceValue
aggregateMetrics(distance, speed)
[API Error]
error
null
estimateDistance(geometry)
calculateFinalScore()
saveCompleteTrip(tripData)
INSERT INTO Trips ...
success
tripID
TripSummaryData
HTTP 200 (TripSummaryData)
Trip Summary Page